

OUTPUTS

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=====
UPLOADED FILES
=====
['.config', 'ir_021_001.wav', 'Second_Part.csv', 'ir_007_003.wav', 'ball_007_002.wav', 'ir_014_004.wav', 'or_014_006.wav', '229.mat', 'or_00
=====
VIBRATION DATA
=====
Used MAT: ['109.mat', '110.mat', '111.mat', '112.mat', '122.mat', '123.mat', '124.mat', '125.mat', '174.mat', '189.mat', '190.mat', '203.mat']
Skipped MAT: []
Raw shape: (6720, 1024, 1)
Feature shape: (6720, 12)
Classes: ['Ball_007' 'Ball_014' 'Ball_021' 'IR_007' 'IR_014' 'IR_021' 'Normal'
'OR_007' 'OR_014' 'OR_021']
=====
CSV DATA
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First CSV: (100000, 11)
Second CSV: (1888178, 26)
Third CSV: (94241, 11)

Final fused shape: (6720, 252)

=====
ACOUSTIC DATA
=====
Total WAV files: 97
Audio feature shape: (97, 31)
Audio classes: ['Ball_007' 'Ball_014' 'Ball_021' 'IR_007' 'IR_014' 'IR_021' 'Normal'
'OR_007' 'OR_014' 'OR_021']

Audio Accuracy: 1.0

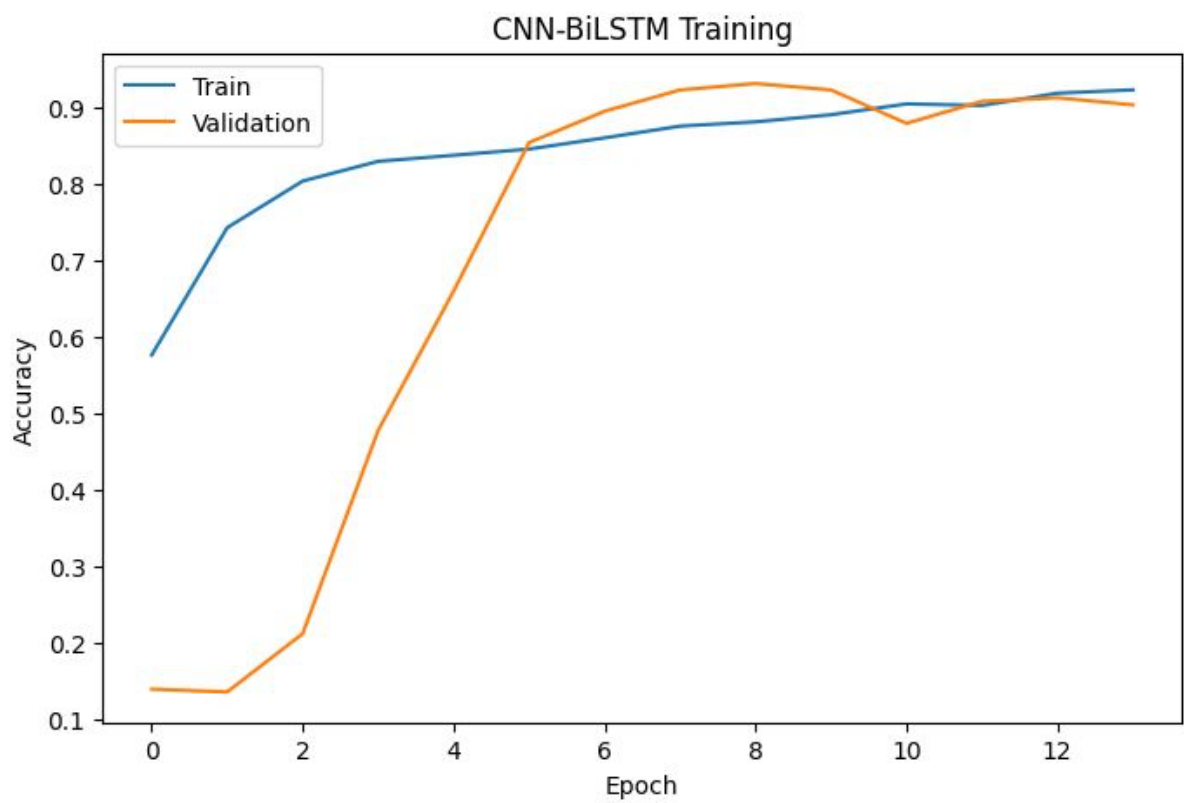
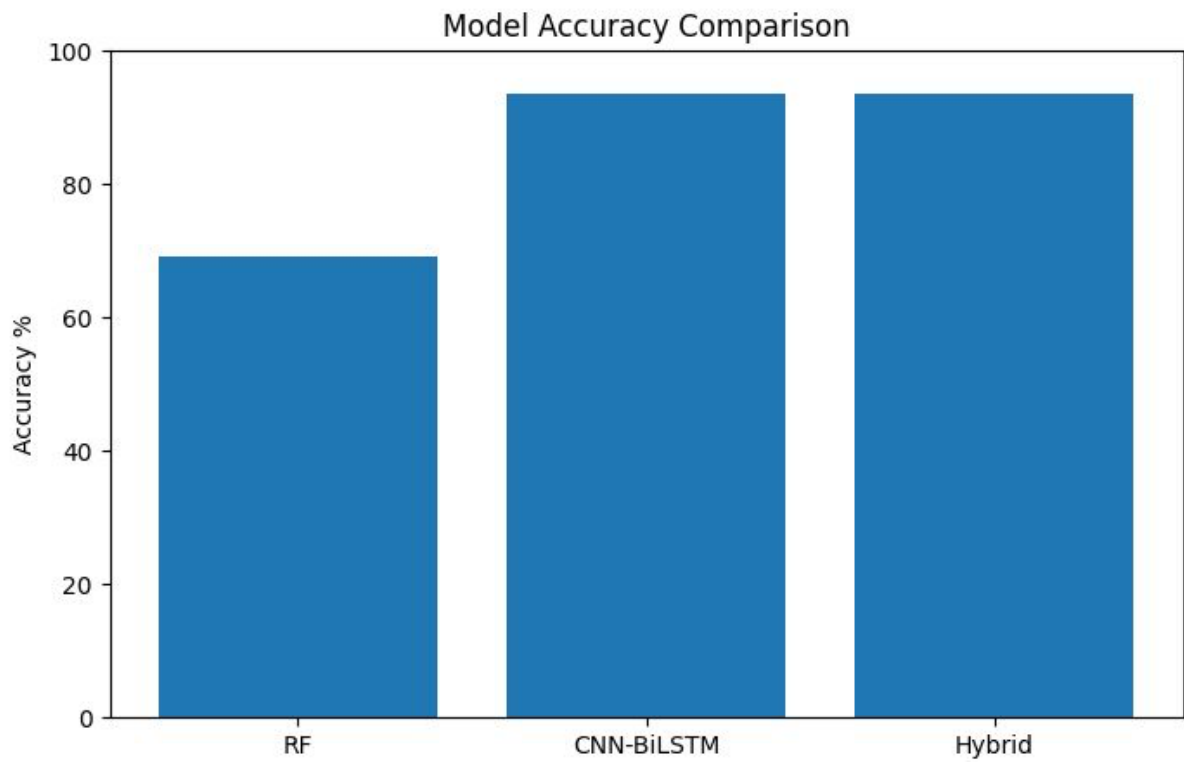
=====
RANDOM FOREST
=====
Accuracy: 0.6912202380952381

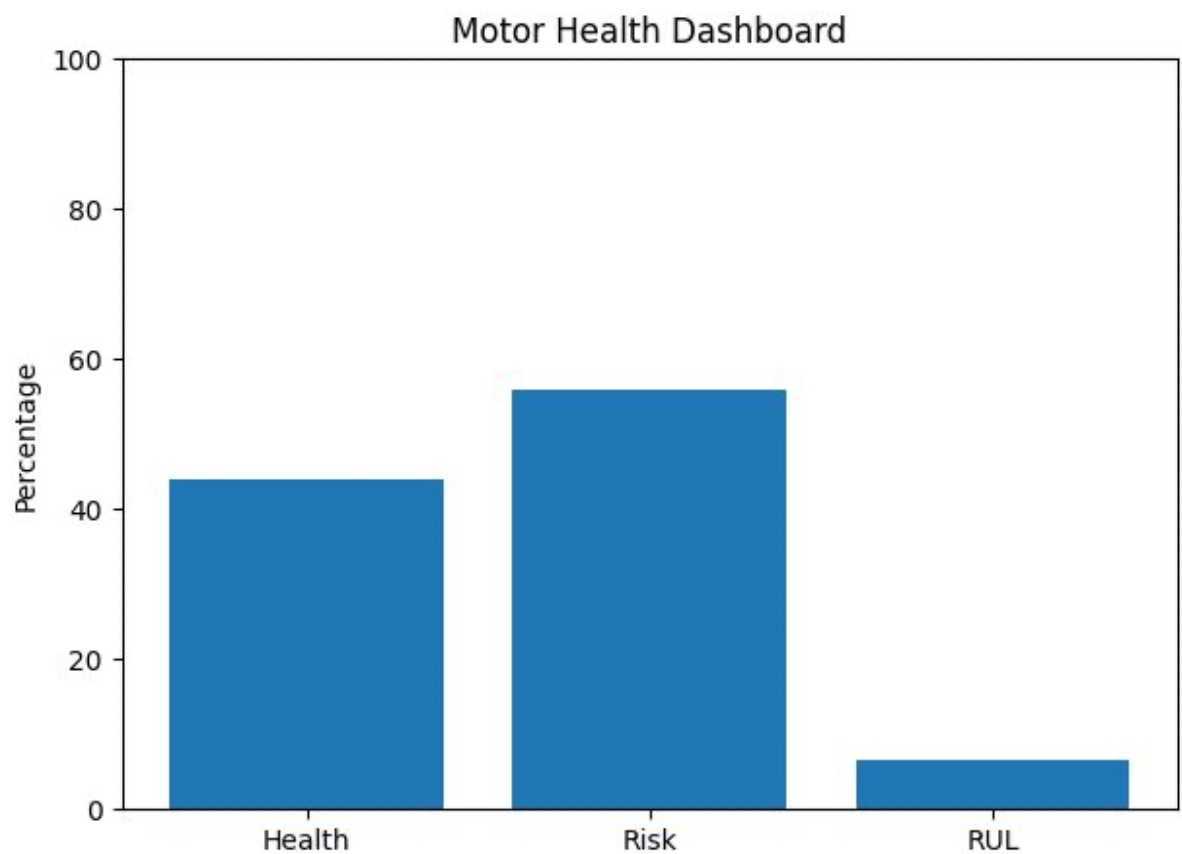
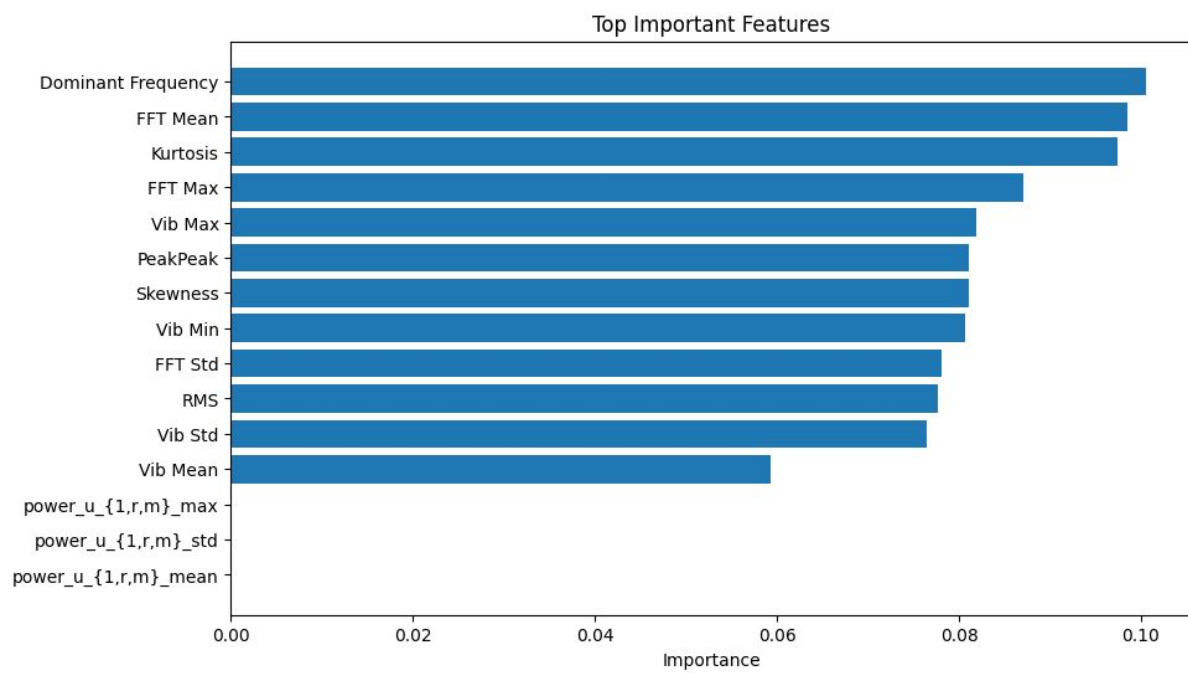
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=====
TRAINING CNN BiLSTM
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Epoch 1/20
135/135 ----- 84s 567ms/step - accuracy: 0.5753 - loss: 1.3417 - val_accuracy: 0.1385 - val_loss: 2.5263 - learning_rate: 0.00
Epoch 2/20
135/135 ----- 81s 561ms/step - accuracy: 0.7419 - loss: 0.7107 - val_accuracy: 0.1348 - val_loss: 3.1381 - learning_rate: 0.00
Epoch 3/20
135/135 ----- 78s 577ms/step - accuracy: 0.8028 - loss: 0.5610 - val_accuracy: 0.2110 - val_loss: 3.3562 - learning_rate: 0.00
Epoch 4/20
135/135 ----- 81s 569ms/step - accuracy: 0.8286 - loss: 0.4713 - val_accuracy: 0.4768 - val_loss: 1.5154 - learning_rate: 0.00
Epoch 5/20
135/135 ----- 83s 579ms/step - accuracy: 0.8365 - loss: 0.4293 - val_accuracy: 0.6589 - val_loss: 0.8657 - learning_rate: 0.00
Epoch 6/20
135/135 ----- 80s 564ms/step - accuracy: 0.8447 - loss: 0.4074 - val_accuracy: 0.8532 - val_loss: 0.3999 - learning_rate: 0.00
Epoch 7/20
135/135 ----- 82s 568ms/step - accuracy: 0.8593 - loss: 0.3832 - val_accuracy: 0.8941 - val_loss: 0.2921 - learning_rate: 0.00
Epoch 8/20
135/135 ----- 82s 570ms/step - accuracy: 0.8747 - loss: 0.3382 - val_accuracy: 0.9219 - val_loss: 0.2129 - learning_rate: 0.00
Epoch 9/20
135/135 ----- 83s 613ms/step - accuracy: 0.8802 - loss: 0.3310 - val_accuracy: 0.9303 - val_loss: 0.2011 - learning_rate: 0.00
Epoch 10/20
135/135 ----- 81s 599ms/step - accuracy: 0.8895 - loss: 0.2827 - val_accuracy: 0.9219 - val_loss: 0.2052 - learning_rate: 0.00
Epoch 11/20
135/135 ----- 80s 583ms/step - accuracy: 0.9037 - loss: 0.2498 - val_accuracy: 0.8783 - val_loss: 0.3179 - learning_rate: 0.00
Epoch 12/20
135/135 ----- 78s 579ms/step - accuracy: 0.9016 - loss: 0.2637 - val_accuracy: 0.9071 - val_loss: 0.2381 - learning_rate: 0.00
Epoch 13/20
135/135 ----- 76s 566ms/step - accuracy: 0.9177 - loss: 0.2219 - val_accuracy: 0.9117 - val_loss: 0.2135 - learning_rate: 5.00
Epoch 14/20
135/135 ----- 82s 612ms/step - accuracy: 0.9221 - loss: 0.2095 - val_accuracy: 0.9024 - val_loss: 0.2459 - learning_rate: 5.00
42/42 ----- 6s 130ms/step

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ADVANCED PERFORMANCE PARAMETERS

Accuracy	: 0.9576
Precision	: 0.9612
Recall	: 0.9576
F1 Score	: 0.9578
Balanced Accuracy	: 0.9574
Matthews Corrcoeff (MCC)	: 0.9509
Cohen Kappa Score	: 0.9505
ROC AUC Score	: 0.9974
Log Loss	: 0.2971
Sensor Fusion Score	: 0.8822
Predictive Maintenance IDX.	: 0.9656
Random Forest Accuracy	: 0.6912
CNN-BiLSTM Accuracy	: 0.9554
Hybrid Ensemble Accuracy	: 0.9576
Acoustic Accuracy	: 1.0000

CONFUSION MATRIX

```
=====
[[ 48  0  0  0  0  0  0  0  0  0  0]
 [ 0 95  0  0  0  0  0  0  0  0  0]
 [ 0 13 167  0  0 10  0  0  0  0  0]
 [ 0  0  0 13  0  0  0  0  0  0  0]
 [ 0  0  0  0 143  0  0  0  0  0  0]
 [ 0 10  4  0  0 80  0  0  1  0  0]
 [ 0  0  0  0  0  0 332  0  0  0  0]
 [ 0  0  0  1  0  0  0 45  2  0  0]
 [ 0  0  0  3  9  0  0  4 174  0  0]
 [ 0  0  0  0  0  0  0  0  0 190  0]]
=====
```

FINAL DASHBOARD

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	Prediction Area	Output
0	Motor Type	3-Phase Induction Motor
1	Best AI Model	CNN-BiLSTM
2	Random Forest Accuracy	69.12%
3	CNN BiLSTM Accuracy	96.58%
4	Hybrid Accuracy	96.35%
5	Actual Condition	Ball_021
6	Predicted Condition	Ball_021
7	Fault Type	Ball Bearing Fault
8	Severity	Severe
9	Confidence	99.55%
10	Machine Health	10.4%
11	Risk Score	89.6%
12	Estimated RUL	1.56
13	Temperature Mean	117.33
14	Temperature Max	14900.0
15	Current Mean	229.13
16	Current Max	3570.26
17	Power Mean	240.6
18	Power Max	14890.96
19	Acoustic Prediction	Ball_007
20	Acoustic Accuracy	100.0%
21	Acoustic Confidence	93.0%

